

THE UNIVERSITY *of* EDINBURGH

MPhys Project Report

Investigating the Nature of Extreme Optical Quasar Variability

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Abstract

It is believed that the radiative activity at the centre of active galaxies is powered by the accretion of matter onto supermassive black holes. Several models over the past 50 years [Pringle and Rees, 1972, Shakura and Sunyaev, 1973, Balbus and Hawley, 1991, Clavel et al., 1992] have attempted to explain their characteristic luminosities and timescales over which we observe them vary. None, however, have successfully predicted the extreme variations we see in modern-day surveys. This Project aims to search for the most extreme examples of quasar variability, in the hopes of testing what makes them different from the general population. The conclusions drawn from this research suggest that the distinctive traits displayed by highly variable quasars are most likely due to environmental influences or a developmental phase that all quasars are anticipated to undergo during their lifespan.

Supervisor: Professor Andy Lawrence

Submission Date: 25/03/2024

Personal statement

The first week was spent deciding which route I wanted the project to go. My options were (i) Finding the statistical properties of the population and correlating it with known properties of the quasars. (ii) Searching for extreme examples of variability to test what makes them different. (iii) Simulate quasars variability using models such as damped random walk (DRW) to try and reproduce their observed variability. My initial instinct was to go searching for extreme examples of variability. To start this off I familiarised myself with the data files provided, understanding the format of the database and the packages I would be using. The next couple of weeks were then spent getting to grips with the IfA computing cluster, Cuillin, the source code editor, VS Code and producing some initial plots. In parallel, I started high level exploration of recent literature within the field.

By mid-October I was fluent in handling the relevant packages along with the full dataset of 500,000 quasars. It was then that I produced my first result showing the anti-correlation between bolometric luminosity and quasar variability. At this point, I made the crucial choice of using the standard deviation (σ) in magnitude as my metric for variability. From here I did my first statistical test on the modality of the distribution of σ. For quite some time there was difficulty implementing the statistics package in Python to obtain the p-values for my results however this was resolved in the background whilst continuing the rest of my analysis.

At the end of October, I began the next stage of my project in searching for extreme variables. We had a couple of brainstorming sessions and eventually came to the idea of comparing the variability (or σ) between two different surveys/epochs from the database. This produced a very interesting result that we wanted to look deeper into. I then defined my own metric for variability by taking the ratio between the σ we see for a quasar and the σ we expect for that given luminosity. I then spent around two weeks fine-tuning the linear regression to make sure the ratio was as accurate as possible. From this, I experimented with some visualisation techniques to see what could be causing the interesting spread of data points we had. This was when I ran into the first hurdle of the project as I realised I hadn't accounted for one epoch having more measurements than the other. After a technical learning curve with some Python packages, I then managed to remove any data that didn't appear in both surveys. This didn't fully solve why our graph looked quite odd/interesting, so I decided to take a deeper look at what was happening in the light curves of the 'unusual' looking quasars.

Once I started looking at the individual light curves of quasars in the region of the graph that looked unusual, I quickly realised that what was causing a lot of spread in the graph was very questionable data points. It was agreed that before I persisted with any more analysis I needed to remove all the quasars that had, what looked like, unphysical data points. As I began sifting through the individual light curves, I realised that an extremely large portion of my 500,000 quasar database had 'questionable' data points. In our last meeting before Christmas break, we discussed my options for how I was going to tackle this problem. Over the break, I experimented with a machine-learning approach to the problem however I ultimately decided that it would be quicker to just remove the data

manually.

For the first two weeks of January, I spent time removing as much questionable data as I could. The more time went on, the bigger the job seemed to get, and upon reflection, it may have actually been quicker to use machine learning. This was also when I started making headway with my report and skilling up on the history and current scene of the research area. Towards the end of January, I began re-doing some of my previous analysis now that I had removed a fairly large portion of my data. Halfway through February I decided to draw a line under removing bad data and just focus on further analysis.

For the second half of February, I set an outline for what I was going to define as an 'extreme variable', and made plots of how these correlated with properties of the quasars. I then spent around two weeks hunting for some values of radio-loudness for the quasars in my catalogue. Some staff members at the Royal Observatory Edinburgh (ROE) kindly pointed me in the right direction to a researcher who gave me access to his newest catalogue. This rounded off my project nicely for another property to correlate with variability. The remainder of the time I then spent writing my report and discussing ideas with my supervisor.

Acknowledgments

I would like to express my sincere gratitude to Professor Andy Lawrence, my project advisor, for his support and invaluable guidance. His extraordinary expertise has been fundamental in shaping my project, and growth as an astrophysicist. I would also like to thank Harry Rendell-Bhatti for his unique database, time and encouragement with my technical challenges. Additionally, I extend my appreciation to Dr. Kenneth Duncan and Bohan Yue for providing me with the data to complete my final analysis.

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1 Introduction

1.1 Motivation

Midway through the 20th century, astronomers began observing some of the most exciting objects to dazzle our skies. Characterised by their strong, broad emission lines and seemingly stellar guise, these objects seized attention when identified as distant radio sources. Prompting profound inquiry, scientists pondered: what could possibly generate such vast amounts of light, that we mistook it as a nearby object?

Active Galactic Nuclei (AGN) was coined as a term to explain the most central regions of some galaxies that exhibit huge luminosities which cannot be due to just starlight alone. Some other energetic process, that is not just star-like nuclear fusion, must be happening to radiate the amount we observe.

Since the early 1970s, theories regarding this physical mechanism have been widely accepted as an accretion disk surrounding a supermassive black hole (SMBH) [Pringle and Rees, 1972][Shakura and Sunyaev, 1973]. During this process, gravitational potential energy is released as in-falling matter from the host galaxy rotationally accretes. At different radii, the disk rotates at different speeds, meaning when neighbouring rings slip past each other, a viscous process induces drag, therefore transferring angular momentum outwards along with local heating.

Several models have been produced over the years in an attempt to explain the physics behind this process - (see section 1.2.2). However, most have stumbled at the hurdle of correctly describing the variability we observe.

The majority of AGN have luminosities which flicker up and down erratically by $\approx 10\%$ on timescales of days, to decades. Over the years, however, more observations are being made of a very small subset that exhibits amplitude changes of up to several orders of magnitude within a matter of weeks. To track these variations, astronomers plot light curves, which show how the brightness of an object varies over time.

As astronomers, we wish to understand what it is that is causing this extreme variability. Is it a physical process that happens in the disk of these specific AGNs? Do they have a distinct property that makes them intrinsically different from the rest of AGN? Or do all AGNs go through periods of rapid variability? Long-term light curves are key to unlocking the answers to these questions, and over the years AGN databases have been growing in length.

For this project, state-of-the-art data, on the brightness of over half a million Quasars (the most luminous subset of AGN) spanning almost 70 years, has been utilised to start characterising the long and short-term behaviour of the most extreme variables. The implications of this are an increased understanding of what is behind the phenomena of the most luminous objects in our universe.

1.2 Background

The following section will concentrate on the theory and physics underlying the foundation of this project. Section 1.2.1 introduces our current understanding AGN and quasars. Following on, Section 1.2.2, sets out the historical timeline of explaining the mechanism behind quasar variability. Finally Section 1.2.3 walks through the statistical models used to quantify the observed variability.

1.2.1 Active Galactic Nuclei

Active Galactic Nuclei (AGN) are the extremely compact regions at the centres of a small portion of galaxies which exhibit a range of properties that starlight alone cannot produce. Luminosities from these regions of less than a parsec in radius have been seen to exceed the amount of starlight produced from a whole galaxy by up to two orders of magnitude. Shortly after a breakthrough in understanding their spectrum in 1963, theories were proposed as to how such vast amounts of energy could be produced in such small volumes. It is now widely accepted that the mechanism powering these objects was a direct result of matter accreting onto a supermassive black hole.

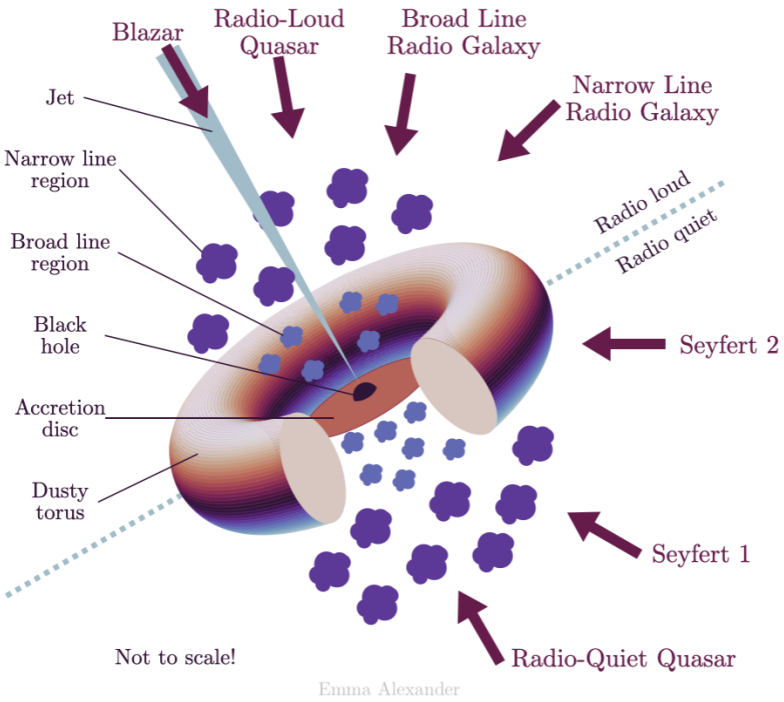


Figure 1: Visual illustration of the unified model of AGN[Alexander, 2022]. Most properties observed in different types of AGN can be explained by the viewing angle of the obsever.

Figure 1 is a visual illustration of the widely accepted unified model of AGN [Urry and Padovani, 1995]. At the centre is a rotating supermassive black hole (SMBH) with a mass typically within the range of $10^6 - 10^{10}$ solar masses ($M_\odot$). Orbiting this SMBH is an accretion disk that harbours sinking gas onto a central plane while maintaining angular momentum[Padovani et al., 2017]. Further out from the accretion disk exists

a 'broad line region' (BLR) which contains rapidly orbiting gas clouds that experience Doppler shifts in their emission lines, hence resulting in broadened lines. Surrounding this BLR is a thick and dusty torus consisting of molecular clouds [Urry and Padovani, 1995] that may obscure the view of the BLR depending on the observer's viewing angle. Outside of the torus, resides the low-density 'narrow line region' (NLR). Some AGN also exhibit extremely powerful, synchrotron emitting relativistic jets [Padovani et al., 2017]. These are outflows which radiate in all wavebands through both synchrotron and inverse Compton scattering events.

Quasars, originally termed Quasi Stellar Objects (QSOs), are a highly luminous category of AGN, with luminosities ranging from $L \approx 10^{38} - 10^{42}$W, often outshining all the stars within their host galaxy [Padovani et al., 2017]. These objects can be characterised by their broad spectral energy distributions (SEDs), emission lines and blue hues. Their most intriguing and elusive aspect, however, is their high variability across the whole electromagnetic (EM) spectrum.

1.2.2 Theoretical Models

Over the years, several theories have been proposed to explain what specifically is happening in accretion disks surrounding SMBHs. We know that when a disk is rotating differentially, a viscous-like process causes friction between neighbouring particles, inevitably leading to heating and then radiation of that energy. This reduces the angular momentum of the particle causing it to drop to lower orbits and therefore drift inwards while transferring its angular momentum outwards to follow conservation.

In 1973, Shakura and Sunyaev [Shakura and Sunyaev, 1973] published *The Standard Accretion Disk Model* which proposed the α parameter to describe the efficiency of this angular momentum transport. Seen in Equation 1, this model assumes a relatively cold ($\approx 10^2 - 10^5 K$), optically thick but geometrically thin disk. If we assume that the viscosity is proportional to the speed of sound, this allows full solutions to the disk structure. However, it does not manage to explain the extremely large luminosities we observe. For this, a much larger increase in viscosity is needed. Normally this is via the addition of turbulence within the gas. In Equation 1, v_t is the turbulence velocity, v_s is the sound velocity and H is the scale height of the disk.

$$\alpha = \frac{v_t}{v_s} + \frac{H^2}{4\pi Q v_s^2} \tag{1}$$

Further theories such as magneto-rotational instabilities (MRI) [Balbus and Hawley, 1991] followed suit to try and solve this problem of additional viscosity. MRI findings suggested that the angular velocity of a conducting fluid in a magnetic field diminishes, with greater distance from the central point of rotation. So in short, it meant that magnetic stresses in the accretion disk produced a viscosity-like effect. This was a triumph in explaining characteristics such as the UV peak in AGN's SEDs, however concerns were still had surrounding the variability of these objects. To explain the large luminosities we observe in the optical using MRI, theory predicts it should be varying on time scales of thousands of years, not the weeks/months that we observed in some cases.

As always in science, however, the crisis was solved with another theory. X-ray reprocessing rescued the problem of short-term, longer wavelength variability by proposing the rapid variations in the far UV, were a result of the inner more central X-ray source, shining on the disk and heating it to re-radiate [Clavel et al., 1992]. Because the X-rays originate from a smaller more central region, it was physically possible for them to vary on much shorter time scales.

With the onset of new large-scale sky surveys in the 21st century though, it became increasingly clear that extreme variability was also occurring in the optical regime, and not just on a handful of occasions. Recent work even suggests that ≈30-50% of quasars vary by large/significant amounts [Rumbaugh et al., 2018]. It is therefore inevitable to question whether the outer optical regions of the accretion disk are experiencing real, physical changes.

There are several different timescales to consider when looking at what physical process is propagating these variations, some of which are more important than others. It is always logical to start by examining the light-travel timescale $t_{lt} = \frac{R}{c}$. This is, by the laws of physics, the shortest timescale we could theoretically see variability on. It was soon concluded though, that this was not the primary timescale of variability, as observations showed timescales much larger than the minimum set by the speed of light. The dynamical timescale $t_{dyn} = \sqrt{\frac{R^3}{GM}}$ describes how quickly we are likely to observe physical changes within the region of the disk. Some say a more realistic timescale is that of sound crossing $t_{snd} = \frac{R}{v_{snd}}$. Finally, the thermal timescale t_{therm} describes the time it would take for energy to disperse within the region.

Table 1: AGN Propagating Timescales [Lawrence, 2016]

Region	t_{lt}	t_{dyn}	t_{snd}	t_{therm}
Optical	1.4 hrs	4.3 hrs	1.3 yrs	18.7 days

The answer to which of these timescales corresponds to the physical process that is happening within AGN is still a topic of debate. However, the collective effort to observe how the flux from these objects changes over time remains the most important key.

1.2.3 Erratic Time Series

As with many processes in nature, the behaviour of quasar optical variability can be understood through the lens of time series analysis. Time series data, representing the sequence of brightness measurements for each quasar over time, often exhibit varying degrees of erratic fluctuations. These fluctuations can be categorized into stochastic, deterministic, or a combination of both components.

Light curves of quasars observed in the optical regime are often modelled using the first-order autoregressive process (AR(1)) - also commonly known as a "damped random walk". In this model, the dynamics are captured by taking each new brightness value as a linear combination of previous values, plus a random term (see Equation 2). This

framework is effective in describing variations within a relatively narrow range, typically around 10-20 per cent. However, it falls short when attempting to explain the immense fluctuations observed in extreme cases, where brightness can vary by over an order of magnitude.

$$x_t = \alpha x_{t-1} + Z_t \tag{2}$$

The AR(1) process, characterised by the alpha (α) parameter typically ranges from 0 to 1. When $\alpha = 0$, it is a purely random process, governed by the distribution of the random term Z_t. When $0 < \alpha < 1$, a flicker noise type sequence is produced, as shown in the top panel of Figure 2. This is what is called a *damped random walk*. When $\alpha = 1$, the well known *random walk* is produced. Once α exceeds 1 however, the sequence rapidly diverges, as illustrated in the bottom left of Figure 2. If we know the form of Equation 2, that quasars and their extreme fluctuations follow, we can not only predict future outbursts but also unravel the mechanism that causes it. This is currently a very important area of research within the AGN community.

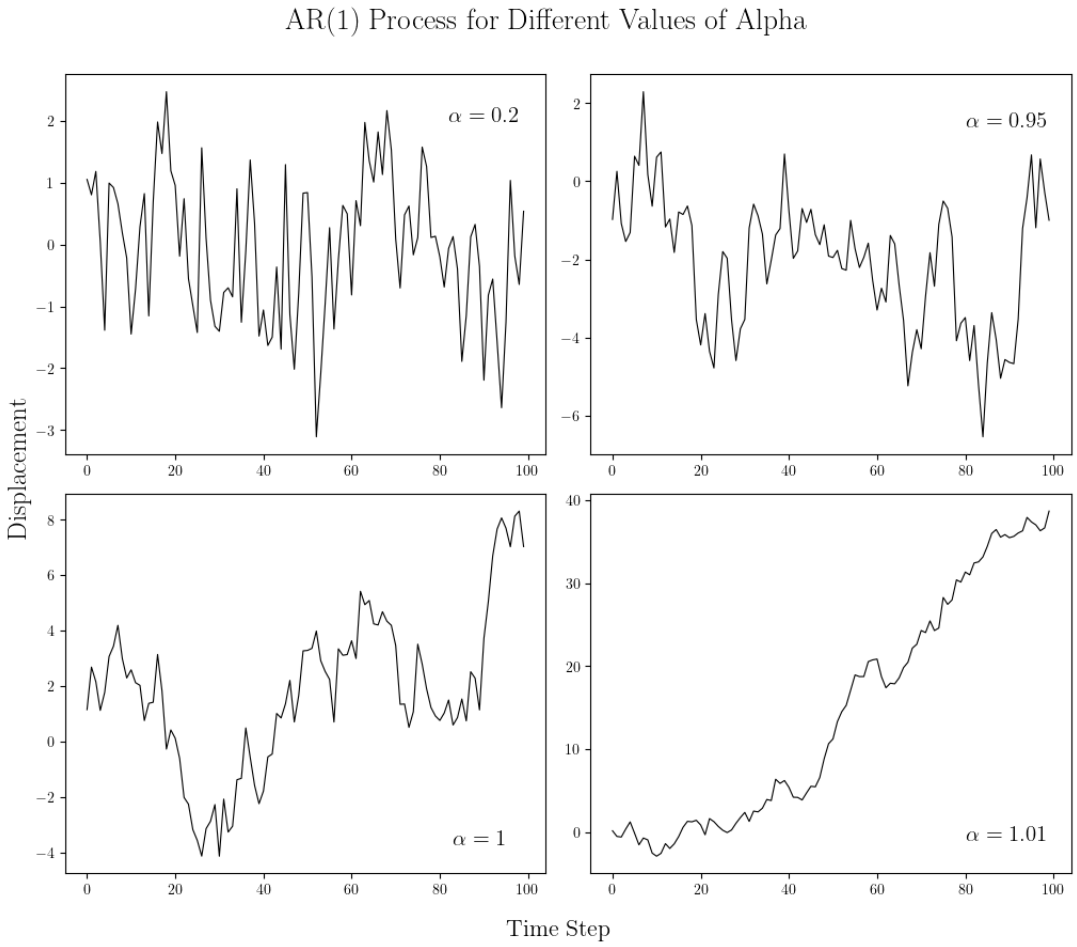


Figure 2: Graphical illustration of the first order Autoregressive process (AR(1)) for different values of α. Top left left and right show different degrees of a damped random walk. The bottom left is a pure random walk, while the bottom right is the rapid divergence after $\alpha > 1$

2 Methods

2.1 Visualising the Data Set

The first step was to begin visualising the data set of half a million quasars. Provided were 100 files, each containing data for 5000 individual quasars. Every quasar had an assigned unique identifier (uid) and Survey ID, several measurements of the magnitude, its error, the filter band (g, r, i) and the Modified Julian Date (MJD) which is the date of observation in the observer frame. For the purposes of this project, the MJD in the rest frame of the quasar was also provided, calculated from Equation 3.

$$\text{mjd}_{\text{rf}} = \frac{\text{mjd}}{1 + \text{z}} \tag{3}$$

Table 2: Example dataframe of the first entry of the dataset. From left to right, quasar ID, Modified Julian Date, Modified Julian Date in the rest frame of the quasar, magnitude, magnitude error, photometric band and survey ID.

uid	mjd	mjd $_{\text{rf}}$	mag	mag_{err}	λ-Band	sid
1	55477.344	16765.592	22.171	0.186	g	7

It is also important to note that the number of measurements for each quasar varied due to different survey coverage. A larger sample size gave the probability of a more accurate representation of individual quasar variability.

As can be seen from Figure 3, the time between observations varied quite dramatically. This can obscure the more detailed and interesting parts of the light curve. Figure 4 zooms into the ZTF observations of Figure 3 to emphasise the level of detail we have in some regions of the time series.

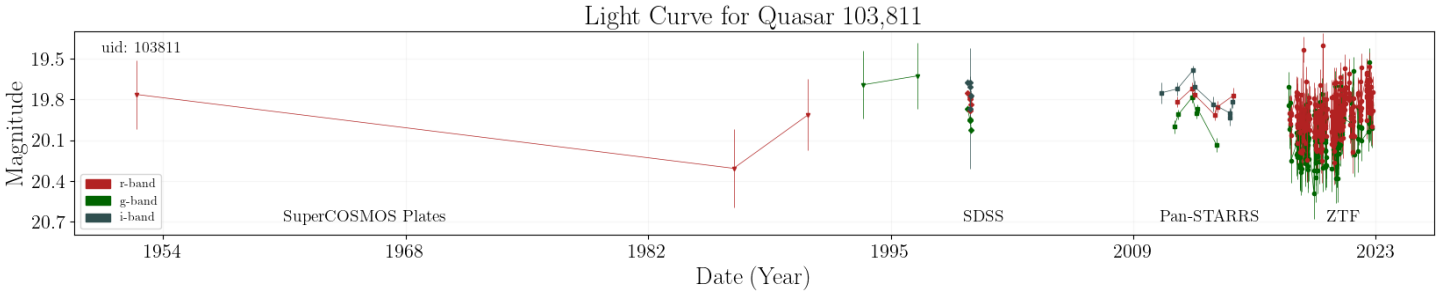


Figure 3: Light curve for quasar 103,811. Spanning from the 1954 Schmidt Telescope SuperCOSMOS plates through to SDSS, Pan-STARRS and ZTF. The different colours represent the different photometric bands observations were made in.

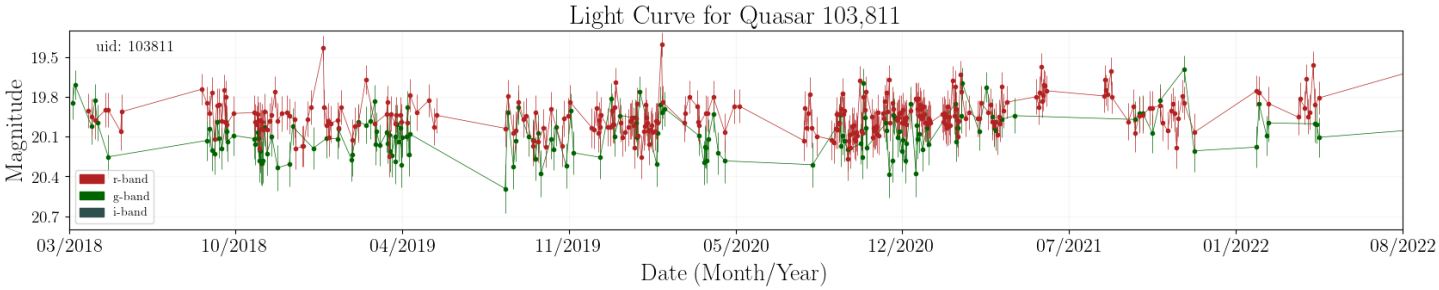


Figure 4: Zoomed in region of quasar 103811 light curve ZTF observations. This emphasises the much finer details of variability happening on short timescales.

To explore the relationship between a quasar's variability and its intrinsic luminosity, Figure 5 was plotted. The standard deviation (σ) of measurements was chosen to be the metric for describing variability. A subtle but present anti-correlation can be seen between variability and bolometric luminosity. It is a well-known relationship, that intrinsically brighter sources vary on less dramatic scales. Examples in literature can be seen here [Berk et al., 2004] [Hook et al., 1994]. It is also very clear however that there is a large spread of data points along the vertical axis. This is interpreted as a large number of quasars which deviate from the typical relationship between variability and bolometric luminosity, e.g. a range of different variability (tending on the larger side) for the same intrinsic luminosity. This immediately prompted attention, as a potential link to our hunt for extreme variables. What was needed next, was a closer look at the distribution of variability within our dataset.

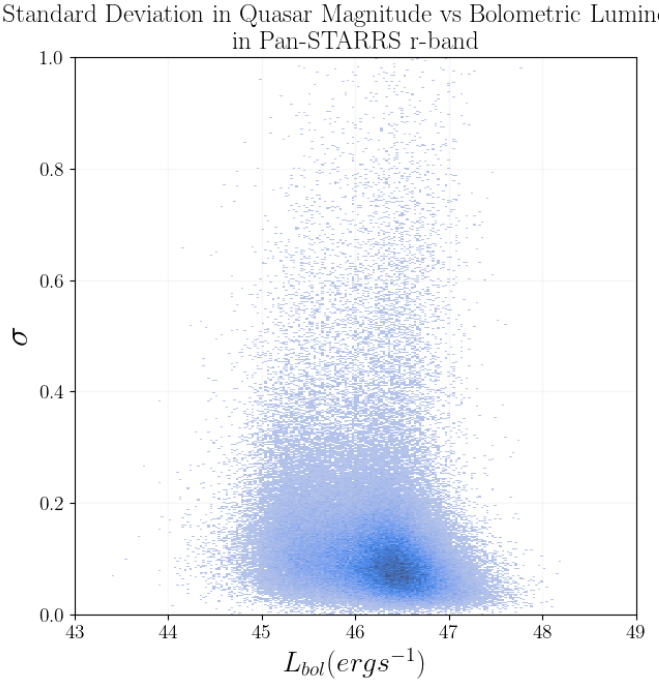


Figure 5: Illustration of the relationship between quasar variability in the r-band and bolometric luminosity in the Pan-STARRS survey. The two properties are seen to be anti-correlated however the large spread of values on the vertical obscures this.

2.2 Determining Modality

As mentioned, the next logical step in the process of analysing the nature of extremely variable quasars was to look at how the variability was distributed. If our theory is that extreme variables are unique in the sense that they are distinct and separate from the general population, we may expect to see more than one mode in a histogram of variability. Perhaps a small increase at the tail of the distribution.

Figure 6 displays the spread of standard deviation (σ) we have for all 500,000 quasars. By eye, one can say this does indeed look unimodal - so our extreme variables fit well into just the tail end of the distribution. To solidify this conclusion, a statistical test was performed to measure the degree of modality. Hartigan's dip test was chosen due to its robust use throughout literature [Hartigan and Hartigan, 1985]. Statistically, it measures how much one needs to change the Cumulative Distribution Function (CDF) of the data, to make it unimodal.

Graphically this can be interpreted using both the probability density function (PDF) and CDF. PDFs measure the probability a variable will take place at x, however, the CDF is the probability a variable will take place less than or equal to x. When a PDF switches from increasing to decreasing, the corresponding CDF will switch from convex to concave, exactly like a functions derivative. This is illustrated in Figure 7.

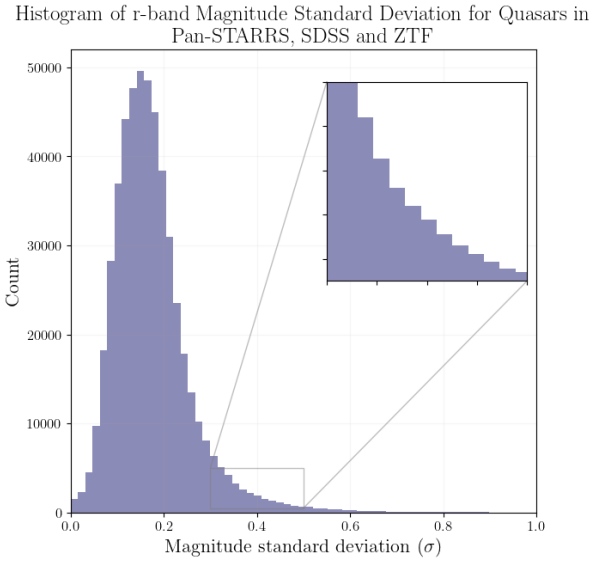


Figure 6: Histogram illustrating the distribution of standard deviation from all 500,000 quasar light curves. Used for determining the populations modality, it is visually clear it is unimodal.

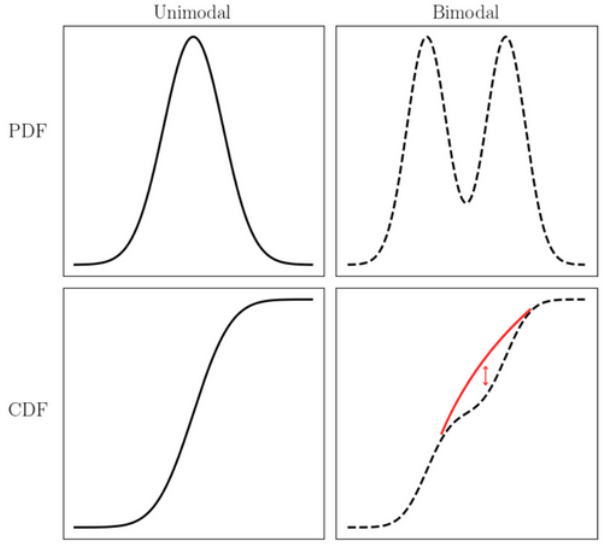


Figure 7: A visual illustration of how modality changes a Probability Density Function (PDF) and corresponding Cumulative Distribution Function (CDF)

Hartigan's dip statistic measures how much one needs to change the CDF to make it unimodal. To implement this statistic on the dataset, the dip test module [Urlus, 2023] was used. This returned a dip statistic of 0.0 with a p-value of 1.0. In J.B. Freeman's paper [Freeman and Dale, 2013], it is quoted that "*HDS values range from 0 to 1 with values less than 0.05 indicating significant modality, and values greater than 0.05 but*

less than 0.10 suggesting bimodality with marginal significance". Here HDS refers to the p-value, which in our case was 1.0. This confirms our assumption of unimodality.

2.3 Does Past Predict Future?

If extreme variables do not deviate from the general population when simply looking at the distribution of σ, do they stand out when looking at their behaviour over time? We know that the general population follows a "damped random walk" model which as previously mentioned, behaves like a random walk on short timescales but achieves a finite variability amplitude as timescales tend toward ∞ [Ivezic and MacLeod, 2014]. But if we predict that the extreme variables deviate from this model, will their σ's stand out when comparing how they change over time?

To begin answering this question, the value of σ for two different surveys was plotted to analyse how the general population evolved. The surveys with the most data points and therefore most information on quasar variability were Pan-STARRS and ZTF. Both of these surveys were also relatively close in time to one another, which was beneficial in the sense that extreme short-term variability was more likely to be captured in action. One might predict the behaviour to follow a linear relationship as, if the brightness of the quasar just flickers up and down stochastically, between two surveys we should expect to see roughly the sample level of variability. What we see instead is interesting.

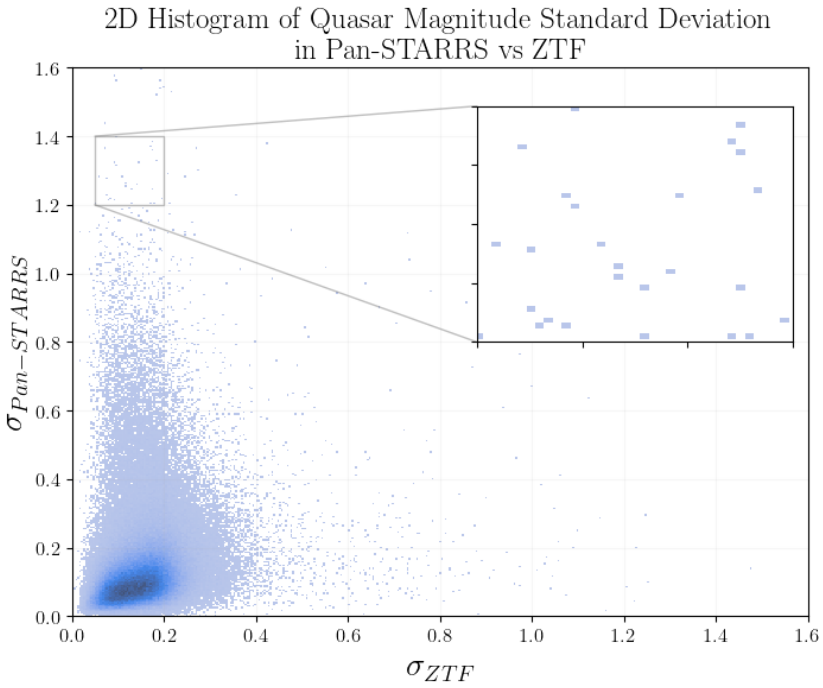


Figure 8: 2D histogram depicting the change in quasar variability between two surveys. The dark region corresponds to the values where most data points sit. The inset aims to emphasise the density of data points which spread out to high values.

Figure 8 has, as expected, a very dense region with linear behaviour. What is striking though, is the large spread of data points particularly vertically but also horizontally. This can be interpreted as a large number of quasars which vary strongly in one survey but not the other. Considering these surveys were less than 10 years apart, this is a striking example of how extreme variability switches on and off for unknown reasons.

To begin investigating these interesting outliers it was decided to define our threshold of what counts as an extreme variable. When looking at Figure 5 of σ vs L_{bol}, we discussed that there is a known relationship and specifically anti-correlation between the two. If we take a linear regression through the dense region which exhibits this relationship, we can define a 'typical' σ or 'level' of variability for a given Luminosity of a quasar. Note that caution was taken in what data to include in the linear regression - we wanted our line of best fit to run through the dense region of anti-correlation, as this is the behaviour we expect from quasars. This meant excluding the large vertical spread of quasars which didn't fit with the general population. The package used to perform the fit was *scipy.stats.linregress* [Jones et al., 2001–] which is a least-squares regression (i.e. minimizing the sum of the squares of the residuals).

$$\sigma_{\text{calc}} = mL_{bol} + c \qquad (4)$$

$$R = \frac{\sigma_{true}}{\sigma_{calc}} \qquad (5)$$

This produced a gradient and a y-intercept from which we could calculate σ_{calc} via Equation 4. If we then took a ratio using Equation 5, we have a new metric to define extreme variability or 'outliers'. Quasars which appeared in the vertical spread of σ vs L because they had unusually high values of σ for that given luminosity, will have a high ratio value (R) in which we can then define a cut-off for 'extreme' variables. Figure 9 shows the relationship between the R-value in both Pan-STARRS and ZTF. This looks very similar to the σ vs σ plot which is to be expected. Now that we have defined our parameter of variability, a closer look at the extreme outliers is needed to get clues as to their nature.

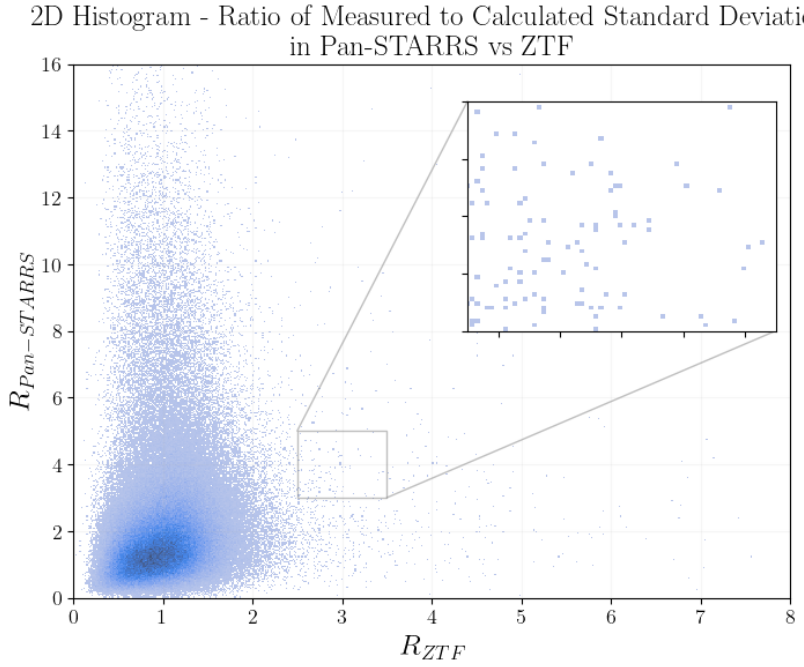


Figure 9: 2D Histogram displaying how the R Ratio of variability changes between the Pan-STARRS and ZTF survey. The inset is used to emphasise the density of data points which spread out to high values.

2.4 Removing Data with Integrity Concerns

As the title of this section suggests, a closer look at the extreme outliers of the R ratio revealed issues with the data. Figure 10 is an example of the types of light curves that were being encountered throughout large values of R ratio. Since there was a considerable number of cases of this, it was decided that no more analysis would be done until they were removed, as this could be skewing our interpretation of how many extreme variables we're seeing, in Pan-STARRS especially.

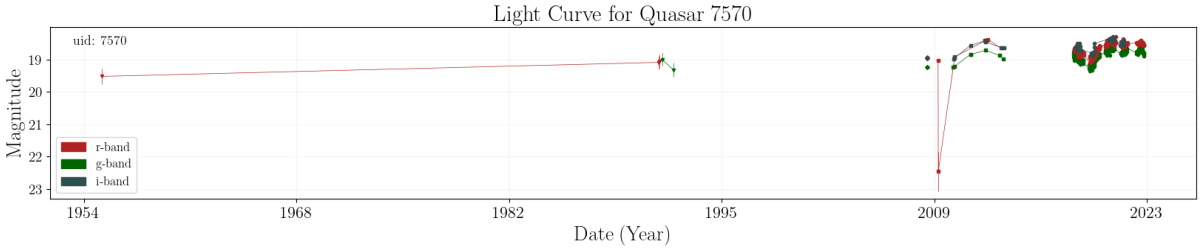


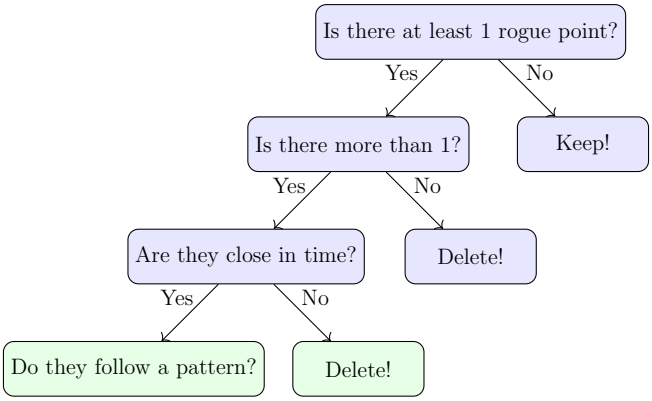
Figure 10: Example light curve of a quasar with a misleading data point. Sometimes it is ambiguous whether a data point could be real, however in this case it is clearly unphysical.

There were two options of approaches to this. The first was to use machine learning, by training an algorithm on what light curves 'should' look like without any rogue data points, so that it would flag up ones such as Figure 10 as a large loss. An Isolation Forest algorithm was considered for this job, as it is known for being a good unsupervised learning technique for anomaly detection, which in our case were rogue and nonphysical data points in our light curve. This is because it attempts to isolate anomalous data points from the beginning of a decision tree by using the average of the predictions.

The second option for filtering out non-physical light curves was to do it by hand, looking at each light curve individually and deciding whether to keep it or not. In the end, the second option was chosen, as because the filtering of suspicious data was so ambiguous, a simple machine learning algorithm would have been too difficult to train on the correct characteristics within the timescale of the project. Choosing the second option though did also come with hurdles. Considering the database had $\approx$ 500 000 quasars, looking at light curves by eye was in itself time-consuming and therefore needed an outline as to how it was chosen whether to keep or delete.

Below is a pictorial representation of how a decision was made. This was a baseline to work from, however, there was a background of questions that could be asked if light curves were ambiguous:

- Does it happen in more than one of the λ-bands? It would be more scientifically convincing if it did.
- Is the order of magnitude it jumps by reasonable?
- Is the lack of supporting data points around this rogue one, just because there weren't as many measurements taken in that survey? The rogue points usually occurred in Pan-STARRS which has much less observational data than ZTF.



As you can see in Figure 11, a significant portion of the vertical spread was from outlier data points. Whether these were incorrect readings from the surveys or were indeed physical (just with a lack of supporting data around it) was up to interpretation. This meant we were left with a continuous distribution of what could be real variability and what was just photometric errors. The possibility of photometric errors could have occurred from the way the database was defined. All observations within a 1-arcsecond radius of the defined coordinates of the quasar were used to produce the light curves. For $\approx 95\%$ of the time this is a good approximation, however, occasionally there will be a star or galaxy within 1 arcsecond of the quasar in the sky, and therefore will be incorporated into the light curve. If this star or galaxy has a very different mean magnitude from the quasar (which is very likely) then random large amplitude changes as seen in Figure 10 could be feasible. Because the database was so large though, we could afford to remove data points that looked false but could have been real, instead of the other way around.

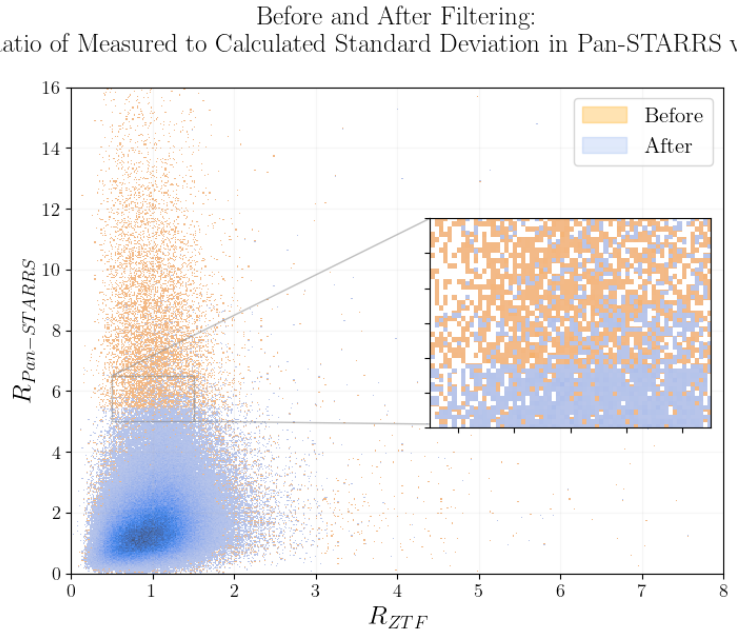


Figure 11: 2D histogram of the R ratio between Pan-STARRS and ZTF, before and after removing light curves with misleading data points. It is clear that this process removed a large portion of the interesting vertical spread, however the inset aims to illustrate that there are still high values of $R_{Pan-STARRS}$ even after the removal.

2.5 Implementing an Artificial Flux Limit

Even after removing quasars and their light curves which had rogue data points, there was still a noticeably larger number of extreme variable quasars on the Pan-STARRS axis. When thinking about what could physically cause this, it was considered that maybe the Pan-STARRS survey could simply see more variability. Upon investigating the flux limit of both surveys, it was found that Pan-STARRS had a flux limit of 24 magnitudes [Kaiser, 2004] while ZTF had a flux limit of 20.6 magnitudes (in the r-band) [Bellm et al., 2018]. This means that Pan-STARRS had the ability to see fainter sources, and therefore may have picked up on smaller/fainter variations in brightness that ZTF simply could not see.

To mitigate this, an artificial flux limit of 20.6 magnitudes was imposed on the Pan-STARRS data set, so that each survey had the same observing capabilities. This meant removing any quasars and their light curves where their maximum value in magnitude within the Pan-STARRS survey was > 20.6. This condition removed 279,194 quasars from the original 504,321 (before filtering out those that did not appear in both surveys or had rogue data points). Once the artificial flux limit was in place, along with light curve removals, 205,203 quasars remained.

As illustrated in Figure 12, implementing the flux limit significantly reduced the vertical spread of variable quasars on the Pan-STARRS axis. Although it still looked like there were more on the vertical axis than on the diagonal or ZTF axis, a more robust approach was needed to quantify how it varies.

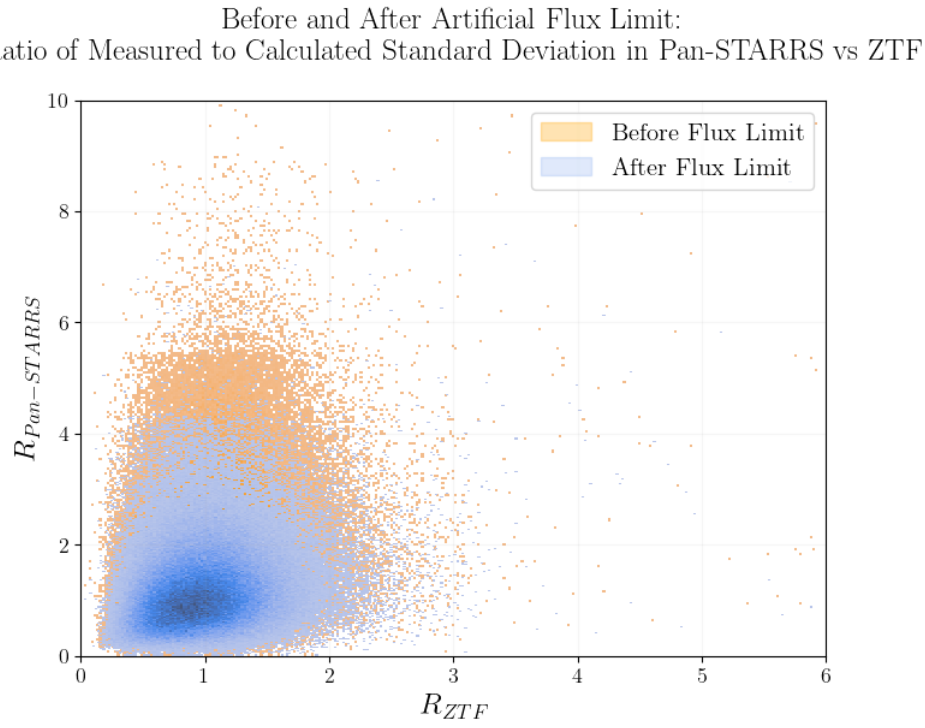


Figure 12: 2D histogram of the R ratio between Pan-STARRS and ZTF, before and after implementing an artificial flux limit on the Pan-STARRS measurements. It is clear again that this process removed a large portion of the interesting vertical spread.

2.6 A Look at Extreme Value Subsamples

Now that there was a corrected sample of quasars and their σ Ratios, we needed a way to quantify the abundance of extreme variables across the different regions of the graph. We define three axes: one horizontal - for quasars which vary strongly in ZTF but not Pan-STARRS, one vertical - for quasars which vary strongly in Pan-STARRS but not ZTF, and one diagonal - for quasars which vary strongly in both. To begin, constraints were implemented to define each sub-sample, as shown in Figure 13.

The cut-off point for an extreme value subsample was desired to be at the Ratio value where the bulk density drops off. This is because we expect our 'extreme variables' to lie outwith the general population. For the ZTF subsample, the constraint was put at the coordinates (2.5,2.5), as it can be seen this is where the general population fades. One may expect the coordinates for the Pan-STARRS subsample to be symmetric with ZTF, however, because the light curves were only filtered down to a Ratio of $R_{Pan-STARRS} = 4.4$ (due to time constraints), the bulk density of data points spreads much higher. As mentioned previously in Section 2.4 this may just be due to continued unreliable data within the light curves that have not yet been filtered out. Nonetheless, the cut-off point for the extreme Pan-STARRS subsample was moved up to this drop off of $R_{Pan-STARRS} = 4.4$ so as not to accidentally include data in our extreme subsample which may be inaccurate. For the diagonal subsample, everything in between the Pan-STARRS and ZTF subsamples was included as by definition, these were the quasars which varied strongly in both surveys.

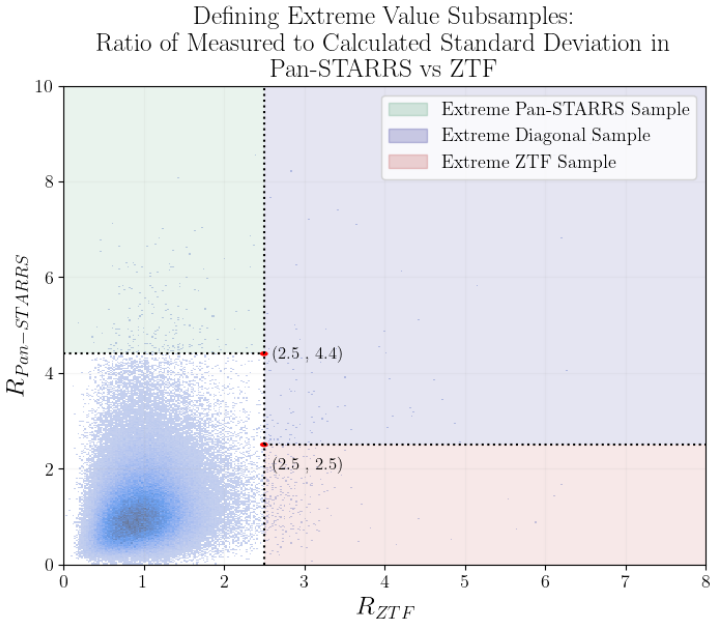
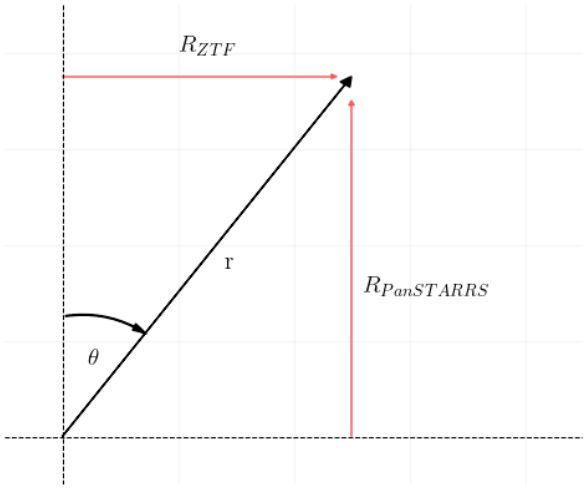


Figure 13: Illustration of how the extreme value subsamples were defined, and where this sit with respect to the dense 'normal' region.

Now that there were three distinct sub-samples defined, a way of determining numerically and visually the density of data points within each, was needed. For this, polar coordinates were employed.

Traditionally, the polar coordinate system measures theta from the x-axis, however for the aid of visually seeing how the densest region on the Pan-STARRS subsample changed while sweeping through into the diagonal and ZTF, $\theta = 0$ was defined as the vertical. This can be seen in Figure 14 and Equations 6 and 7 with the 'y' coordinate defined as R_{ZTF} and the 'x' coordinate defined as $R_{\text{Pan-STARRS}}$.



$$r = \sqrt{R_P^2 + R_Z^2} \tag{6}$$

$$\theta = \arctan 2 \left(\frac{R_{\text{ZTF}}}{R_{\text{Pan-STARRS}}} \right) \tag{7}$$

Figure 14: Pictorial representation of the way we switched from Cartesian to Polar Coordinates. In particular, that we defined our origin of θ as being the vertical axis.

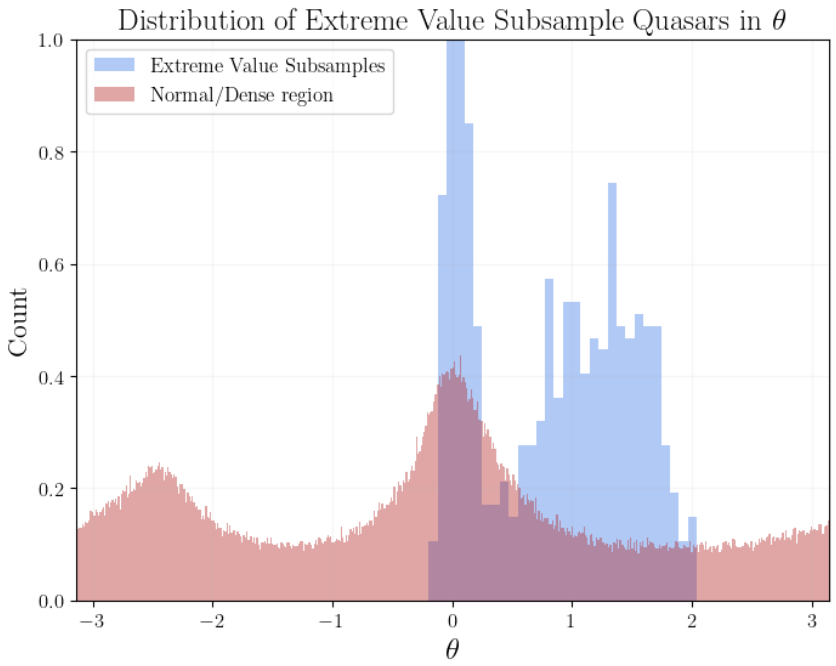


Figure 15: Histogram displaying how the density of data points in the R ratio plot, varies as we sweep through the subsamples. As expected there is a large peak in the Pan-STARRS subsample before dropping off and gradually rising again through the diagonal, to ZTF.

Figure 15 shows how the number of quasars with high variability ratios changes between the extreme subsamples. The most prominent peak between $-0.2 < \theta < 0.2$ represents the Pan-STARRS subsample. This peak is expected due to the larger spread of more variable quasars in the Pan-STARRS survey. Between $0.2 < \theta < 0.6$, there is a steep dip. This represents the region between the Pan-STARRS and diagonal subsample which contain far fewer highly variable quasars. For $\theta > 0.6$ there is a slower increase in the number of quasars, as well as no second dip corresponding to the change between the diagonal and ZTF subsamples. This is interpreted as the density of highly variable quasars plateauing when moving towards lower variability in the Pan-STARRS survey. The distribution of θ for the dense region of Figures 13 and 12 (or what has been classified as 'normal' varying quasars) has been over-plotted as well to compare how the shapes of the distributions vary. One may expect the distribution of this to be flat, if for 'normal' quasars, there is no one-preferred subsample, e.g. quasars are just as likely to vary more in one survey than they are in the other or both. It may also be expected that there should naturally be a peak at $\approx 0.8 rad$ if quasars vary consistently throughout time.

However, again the peak is at $\theta = 0$ due to the excess of quasars seen varying more in Pan-STARRS. It must be considered though that many of these may contain unreliable data points that have not been filtered out yet. No second peak is observed beyond $\frac{\pi}{4}$ which means there is not a higher number of more variable quasars in the ZTF survey for the general population.

2.7 Properties of Extreme Variables:

To further investigate whether extremely variable quasars are intrinsically different in some way, comparisons of properties to the general population were made.

2.7.1 Light Curve Trends

As illustrated in Figure 16, when looking at quasars with a high σ Ratio, the light curve presents as relatively normal, with only slight fluctuations, until it suddenly transitions to extreme variability. Could this mean that in these quasars, some physical process is being turned on in the time between surveys? When looking at quasars with high σ Ratios in both surveys however (i.e. in the Diagonal subsample), large and rapid variations in magnitude are seen consistently throughout multiple surveys. To enhance the scale on which these variations are happening, an example of what a 'typical' quasar light curve looks like over 70 years is displayed at the bottom of Figure 16. As you can see, small-scale fluctuations do occur over the years, as this is one of the defining characteristics of AGN, however nothing close to what we observe for our extreme variables sub-sample.

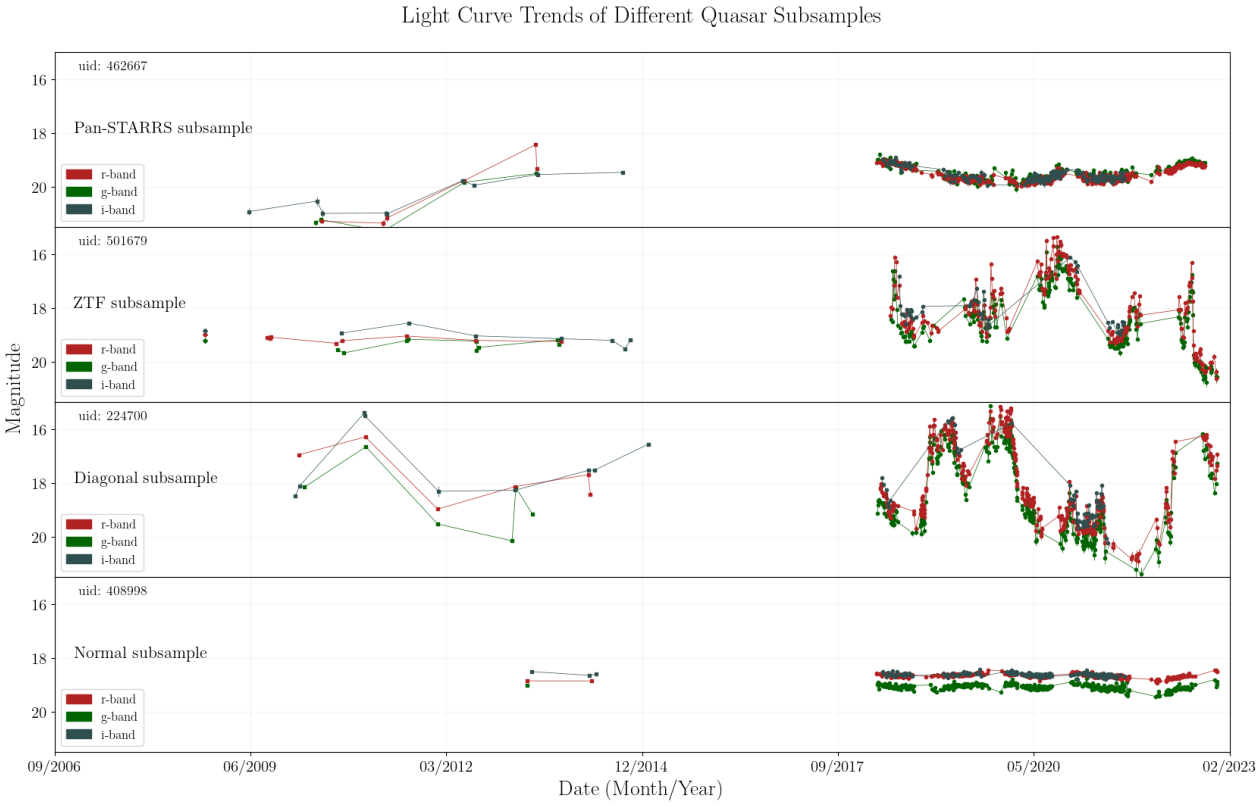


Figure 16: A series of lightcurves chosen from different extreme value subsamples within the R ratio plot and zoomed to better illustrate the variability in the later surveys. This is to illustrate how the pattern of variability changes between the three. A fourth lightcurve chosen from the 'dense'/ 'normal' region is also displayed at the bottom to emphasise the scale at which the extreme variability is occurring on. As can be seen, the Pan-STARRS extreme value subsample has long and large outburst during that survey, before virtually returning to 'normal' in ZTF. The ZTF extreme value subsamples has the opposite, a mostly calm lightcurve during Pan-STARRS before exploding into variability in ZTF. The diagonal subsample has extreme fluctuations in both surveys while the 'normal' light curve exhibits an almost flat trend.

2.7.2 Black Hole Mass

An intrinsic property of an AGN is the mass of the black hole that matter is accreting onto. In the catalogue of properties used so far from SDSS data release 16 (DR16) [Wu and Shen, 2022], black hole mass was fiducial single-epoch. One may presume that the more matter that falls onto the SMBH (and therefore the higher M_{BH}), the more particles and dust will heat up and produce the radiation and luminosity that we see. The anti-correlation between variability amplitude and Luminosity has been established for quite some time as discussed in Section 2.1 [Berk et al., 2004] [Hook et al., 1994]. So perhaps the more highly variable quasars should have larger black hole masses.

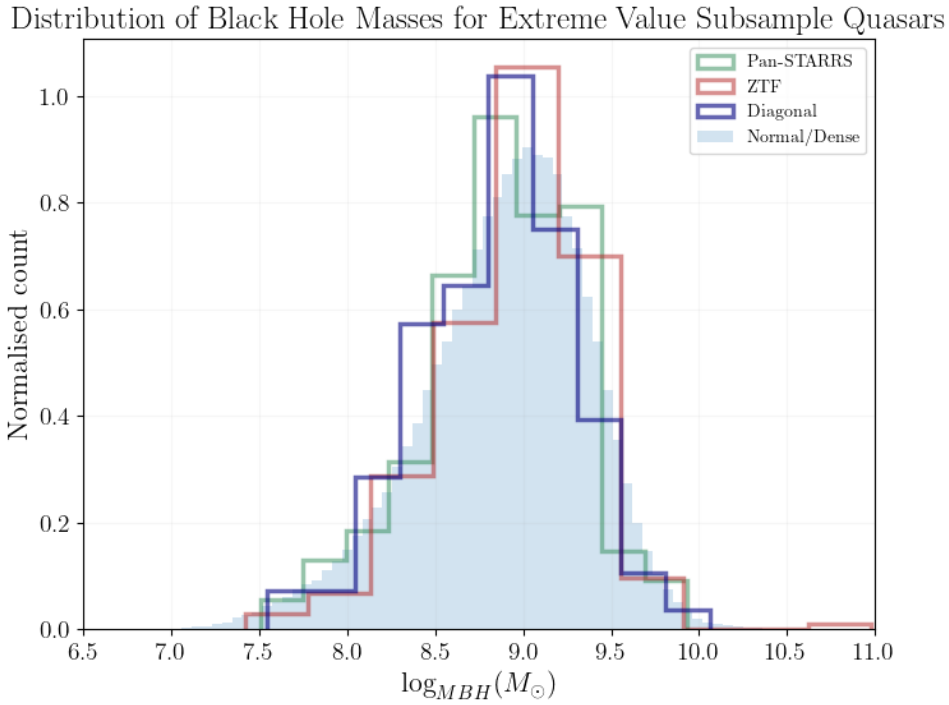


Figure 17: Histogram displaying the distribution of black hole masses for different subsamples. The shapes of each population look almost identical, meaning there is no indication of a correlation between variability and black hole mass.

As illustrated in Figure 17 however, this is not the case. The distribution of black hole masses for the highly variable quasars looks almost identical to the distribution for more 'normal' ones. Throughout the literature, this result has remained elusive, with different studies producing contradicting results [Arévalo et al., 2023]. However, a recent study published in 2023, claims the degree of optical variability decreases with black hole mass on short timescales [Arévalo et al., 2023]. In that study, the key to establishing the relationship was to look at the quasar light curves on much shorter timescales ($\approx$ 30 days). The length of light curves used to establish the degree of variability in this project though was roughly 70 years, so perhaps any relationship between black hole mass and variability has been skewed by the length of observations.

2.7.3 Eddington Ratio

Another intrinsic property of AGN and its host SMBH is the Eddington ratio. The definition of the Eddington ratio is shown in Equation 8 which essentially computes the proportion of the bolometric luminosity to the Eddington luminosity. The Eddington luminosity or 'limit' is the maximum luminosity the AGN can reach while in hydrostatic equilibrium [Peterson, 1997] or when the gravitational attraction of the black hole balances with the radiation pressure produced from the accretion of infalling matter. This can be calculated for an object via the properties in Equation 9. In the DR16 catalogue used for this project, L_{Edd} was calculated using the fiducial single-epoch BH mass as seen in the previous section.

$$R_{Edd} = \frac{L_{bol}}{L_{Edd}} \tag{8}$$

$$L_{Edd} = \frac{4\pi G M m_p c}{\sigma_T} \tag{9}$$

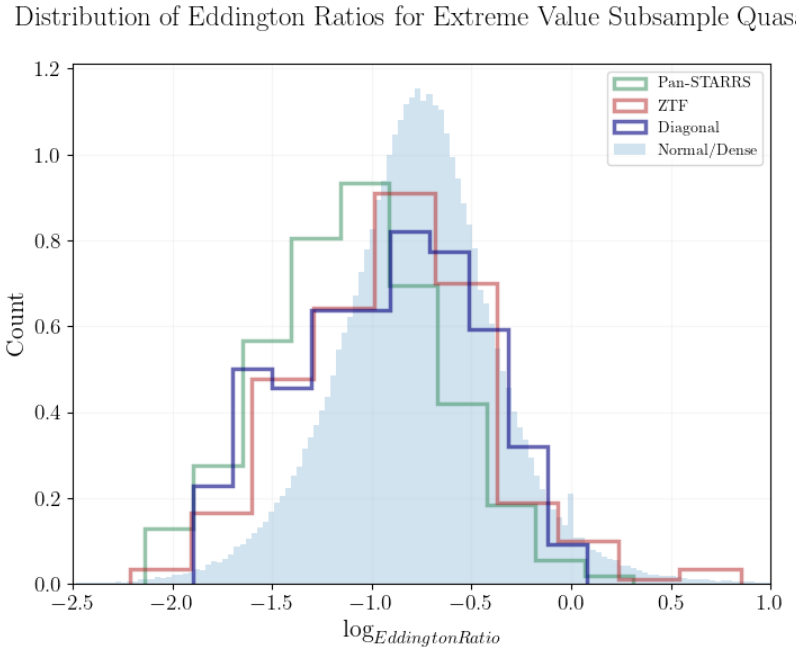


Figure 18: Histogram displaying the distribution of Eddington ratios for different subsamples. All extreme value subsamples deviate from the 'normal' population towards lower values of Eddington ratio.

Figure 18 shows a much more interesting deviation for highly variable quasars from the general distribution of normal quasars. All extreme subsample quasars tend to lower Eddington ratios compared to the normal population, especially those which vary highly in the Pan-STARRS survey. Performing a two-sample Kolmogorov–Smirnov (KS) test on the overall distribution yields the following results in Table 3. This test, in short, determines whether two samples of data came from the same distribution by comparing

their cumulative distribution functions (CDFs). The statistic represents the maximum difference between the two CDFs, while the p-value gives the probability of observing the distributions as far apart as given, assuming they are sampled from the same population.

Table 3: Comparison of the KS-test results for the Eddington ratio distribution.

Axis to compare to general population	Statistic	P-value
Pan-STARRS	0.344	4.2×10^{-24}
ZTF	0.144	8.6×10^{-6}
Diagonal	0.192	4.6×10^{-4}

The most significant result is that of the Pan-STARRS highly variable quasars which have an astoundingly low probability of coming from the same distribution as the 'normal' quasars. This is probably in part, because of the size of the dataset, as the probability of observing deviations from the general distribution becomes less and less likely with increasing population. However, it is a result that is hard to ignore, especially with the other two sub-samples of highly variable quasars following the same trend.

Some previous research has pointed towards there being an inverse relationship between optical variability and Eddington ratio [Wilhite et al., 2008] [Sánchez-Sáez et al., 2018] [MacLeod et al., 2010] [Simm et al., 2016]. That is, quasars with lower Eddington ratios are more likely to have higher variability. Physically this may be interpreted as a scenario where variability in the optical region is somehow tied to variation in the accretion rate.

2.7.4 Luminosity

Discussed previously was the well-established anti-correlation between the amplitude of variability and bolometric luminosity i.e. intrinsically brighter quasars vary on smaller scales. Figure 19 shows how the number of quasars of different variabilities is distributed across L_{Bol}.

At first glance, the distribution for 'normal' quasars has a very interesting shape - asymmetric, and the potential for bimodality. While the normal population dips in numbers towards lower luminosities, the highly variable Pan-STARRS and Diagonal quasar peak at lower luminosities!

The SDSS, like all surveys, was flux-limited (e.g. you can see everything down to a certain brightness) however in nature there are actually many more faint objects than bright objects. The combination of these two effects makes the distribution of different luminosities tend to accumulate near the middle, which is what we see in Figure 19.

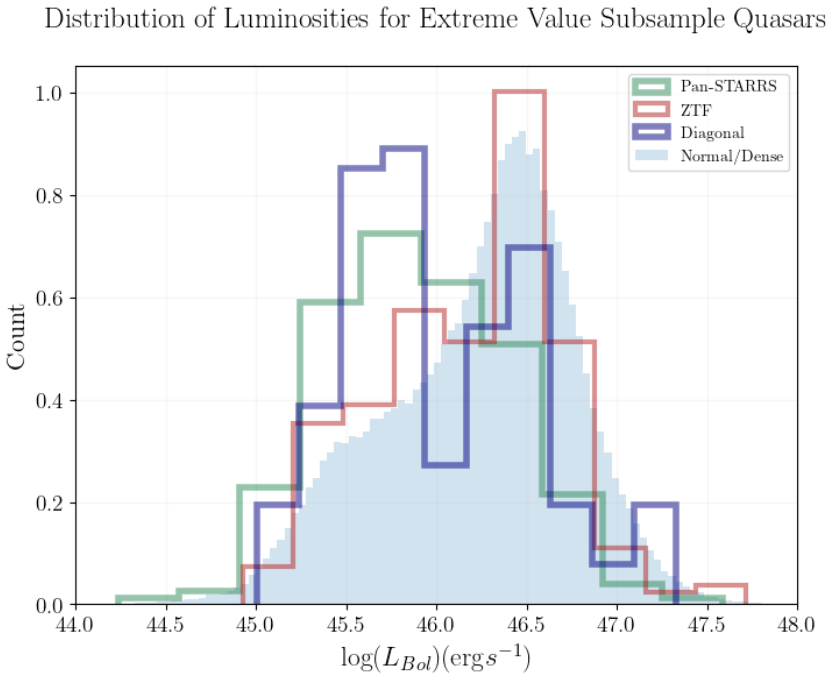


Figure 19: Histogram displaying the distribution of luminosities for different subsamples. Most noticeably, the Pan-STARRS and diagonal subsamples deviate significantly from the 'normal' population, while for some reason, ZTF does not.

What is interesting about the deviations in distributions is the fact that technically there was a correction for the relationship between variability amplitude and luminosity. Back in Section 2.3, the σ Ratio was calculated by dividing the observed standard deviation by the standard deviation we'd 'expect' for that given luminosity. This means that the more erratic lower luminosity quasars should have been leveled out with the rest of the population. From Figure 19 though it is very clear that the highly variable quasars, particularly found in Pan-STARRS or both with ZTF, have intrinsically lower luminosities.

Table 4: Comparison of the KS-test results for the bolometric luminosity distribution

Axis to compare to general population	Statistic	P-value
Pan-STARRS	0.316	2.6×10^{-20}
ZTF	0.0822	0.0357
Diagonal	0.270	1.3×10^{-7}

Similar to the previous section, a two-sample KS test was performed to determine the probability of observing the extreme-axes distributions, assuming they are sampled from the normal quasar distribution. A table of these results is shown above. This may mean that there is an additional effect that lower-power quasars are more likely to have extreme outbursts, which is somehow not corrected by doing the general σ Ratio.

2.7.5 Radio Loudness

One particularly interesting property of AGN is their radio emission. According to the unified model, an observer can see a quasar as either radio-Loud or radio-quiet [Wilson and Colbert, 1995]. Radio-loud objects are attributed to galaxies with extremely large relativistic radio jets, and lobes which the jet inflates [Hardcastle et al., 2019]. Consequently, synchrotron radiation from these regions dominates the radio emission which we observe. Radio-quiet objects, on the other hand, have an absence of these jets, which in turn allow detection of weaker radio emission from other mechanisms such as star formation and free-free emission [Panessa et al., 2019]. The 'radio-loudness' of AGNs is historically defined as the ratio of radio flux to optical flux [Kellermann et al., 1989], however, the specific wavelength in which these are made varies from paper to paper. Some researchers also suggest that the radio-loudest quasars have systematically larger variability amplitudes by about 30 [MacLeod et al., 2010].

The catalogue of properties used so far for this project was the SDSS data release 16 (DR16) [Wu and Shen, 2022] however sufficient measurements to create the ratio were not available. The SDSS DR7 catalogue contained radio-loud properties which were matched from radio sources in the FIRST survey [White et al., 1997], defined as Equation 10. The cross-over of quasars in the DR7 catalogue and quasars used in this project though was extremely small (≈ 200) and therefore would not have given an accurate picture of any relationship between radio-loudness and variability.

$$RL = \frac{f_{\text{6cm}}}{f_{2500\mathring{A}}} \tag{10}$$

Further enquiry was made to obtain a catalogue of LOFAR (Low-Frequency Array) measurements for SDSS DR16 quasars [Yue et al., 2024]. In this sample, there were a lot more cross-overs of quasar measurements which meant a better statistical approach could be taken to assess the relationship between radio-loudness and variability. Here, the radio loudness was defined as Equation 11. Upon an initial sanity check with calculations from this catalogue, it was realised that the RL values were incorrect. The provider made modifications for a new radio-loud value defined in Equation 12. This is something more commonly used in previous LOFAR papers where L_{150} is the LOTSS radio luminosity and $L_{\text{i-band}}$ is the rest-frame SDSS i-band luminosity. Traditionally by this definition, radio-loud quasars are defined as being $RL > 1$ [Gürkan et al., 2019], however as you can see in Figure 20, there is a second mode after $RL = 2$. So for the purpose of this project, quasars which had an $RL \geq 2$ were considered 'radio-loud'.

$$RL = \frac{L_{\text{150MHz}}}{L_{2500\mathring{A}}} \tag{11}$$

$$RL = \log_{10}\left(\frac{L_{\text{150MHz}}}{L_{\text{i-band}}}\right) \tag{12}$$

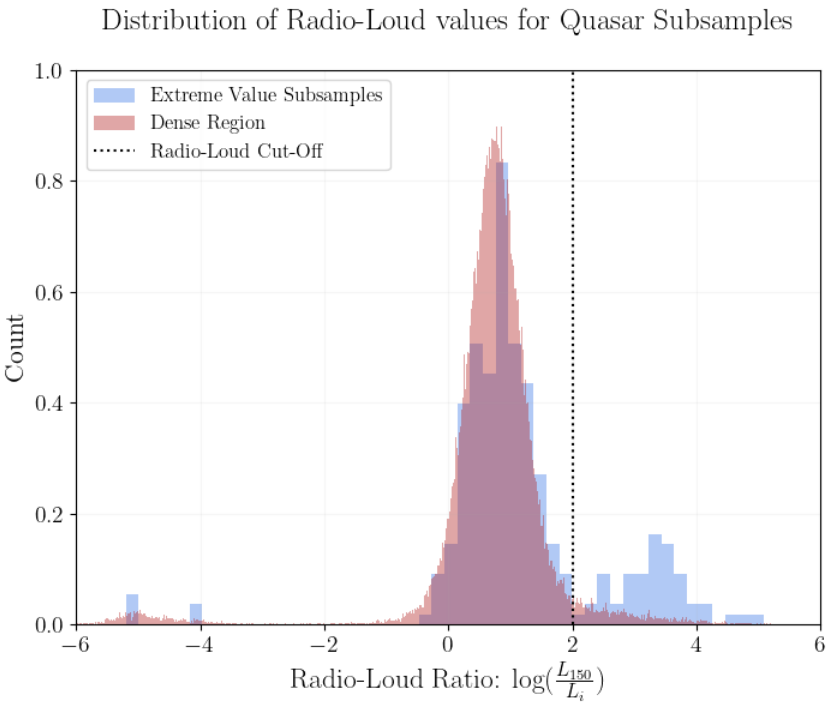


Figure 20: Histogram displaying the distribution of radio loudness for 'extreme' quasars vs 'normal' quasars. There is a fairly prominent second peak in the distribution of highly variable quasars which may indicate a relationship between variability and radio loudness

When performing Hartigans Dip Test of multimodality [Hartigan and Hartigan, 1985] on this distribution (explained in Section 2.2), a dip-value of 0.02 and p-value of 0.51 were obtained. Although this does not qualify as any bi-modality being statistically significant, this is most likely due to the size of the dataset being small (269 quasars).

A further look at where these RL quasars sat on the original ratio plot of $R_{Pan-STARRS}$ vs R_{ZTF} was then made in order to gauge whether there was a correlation between a quasar being radio-loud, and the pattern of its variability. This is shown in Figure 21. It was originally thought that because quasars which lay in the diagonal subsample were consistently more variable and erratic, perhaps they would be the ones most likely to be radio-loud. However as you can see in Figure 21, there is a large spread of locations on the graph, i.e. not just on the diagonal. However, something interesting that pops out is that there seems to be half as many radio-loud quasars that are highly variable in Pan-STARRS but not in ZTF than there are in vice versa. But with the relatively small amount of data at hand, the hypothesis that radio-loud quasars are only the ones which have large and prolonged variability does not seem to be true.

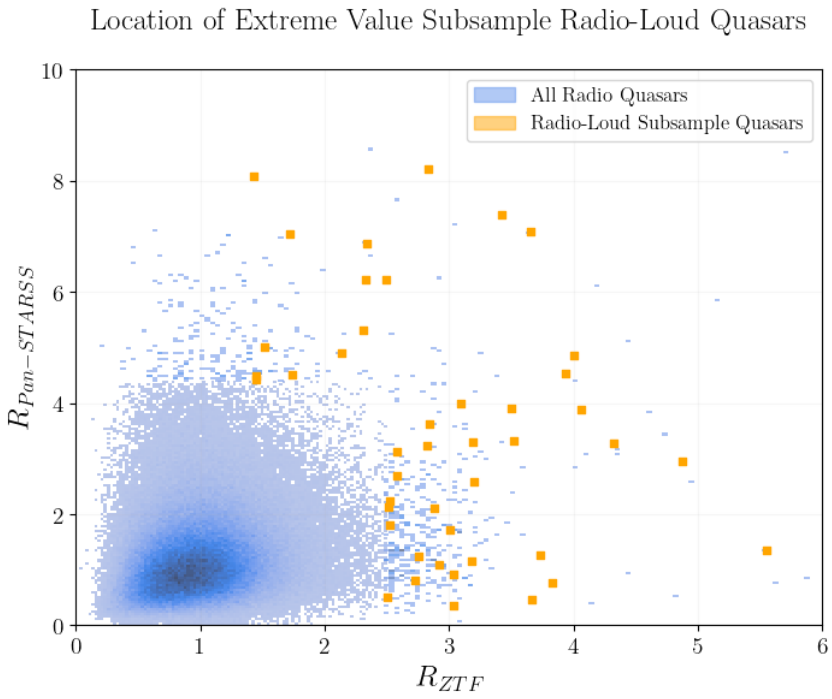


Figure 21: 2D histogram of the R ratio between Pan-STARRS and ZTF and the location of "radio-loud" quasars. The distribution of radio-loud subsample quasars seems fairly random, with no specific region harbouring the majority of data points.

3 Summary of Key Results

To summarise the key findings of this project, the following has been established:

The distribution of σ in the project sample is not bimodal.
Originally it was thought that if highly variable quasars were physically different objects in some way from the general population, they would stand out in their standard deviations (σ). Visually and statistically this was not the case though.

A high proportion of the project sample was distorted by data with integrity concerns.
A large proportion of data with high Pan-STARRS σ values were a result of data points on light curves which had questionable accuracy. Without any supporting surrounding data points or trends, these light curves had to be removed from the sample in order to uphold the integrity of the project analysis.

Statistically there were more highly variable quasars in the Pan-STARRS survey.
Even after removing unreliable quasar light curves and implementing an artificial flux limit, there was still a greater number of highly variable quasars that were observed in Pan-STARRS than ZTF, or both.

The length of highly variable outbursts remains elusive.
Quasars which are flagged up as highly variable in both surveys look to have more long and gradual outbursts, while quasars which are highly variable in only one survey look to have more short and erratic outbursts. Without more detailed observations over long periods of time though, any conclusions about why this happens cannot be made.

There is no observed correlation between black hole mass and variability.
The highly variable quasars from our sample did not show much deviation from the general distribution of black hole masses.

There looks to be a slight anti-correlation between the Eddington ratio and variability.
All sub-samples of highly variable quasars showed the probability of significant deviation from the general population when comparing Eddington ratios, especially Pan-STARRS.

The expected anti-correlation between luminosity and variability was reproduced, with some additional relationship showing to unaccounted for.
Consistent with the literature, the sample produced the anti-correlation between bolometric luminosity and variability. However, even after correcting for this, all sub-samples of highly variable quasars showed significant deviation from the general population. This implies there was an additional effect which wasn't accounted for in the σ ratio procedure.

There is a second mode in the distribution of radio-loudness for highly variable quasars.
Statistically there were more highly variable quasars which were radio-loud when compared to the general population. There was no preferred location on the σ ratio plot though meaning it wasn't clear whether there was a preferred timescale of variability for highly variable radio-loud quasars.

4 Discussion

This investigation into extreme examples of optical quasar variability has provided valuable insights into what physical processes may be governing these interesting objects. This discussion aims to contextualise the key findings of this study by linking them with the broader field of quasar variability research and outline any avenues for potential future work.

4.1 Possible Mechanisms for Large Amplitude Changes

Several physical processes and phenomena have been proposed and discussed extensively throughout literature as being possible causes for extreme variability in quasars. Broadly speaking, they can be split into two categories (i) an internal/intrinsic event which causes substantial physical changes within the quasar itself. Or (ii) external/environmental causes which are a result of how we observe the quasars. The following are examples:

Accretion Disk Instabilities
The accretion disks surrounding SMBHs are hypothesised to undergo both thermal and viscous forms of instabilities. The former may occur due to the interchange between cooling and heating processes within accretion disks. Any imbalances may trigger perturbations in temperature/density profiles which in turn will lead to fluctuations in luminosity. The thermal timescale (t_{therm}) was discussed previously in the introduction as a timescale to explain how long it takes for energy to dissipate within the disk thermally. In observations, this seems to match the variability we would expect to see from the inner disk [Lawrence, 2016]. Viscous instabilities, on the other hand, arise due to the interaction between neighbouring layers of material within the accretion disk. This may include the previously discussed Magneto-Rotational-Instability (MRI), which occurs when magnetic fields couple with rotating gas within the disk [Balbus and Hawley, 1991], or thermal-viscous instabilities which occur from variations in the disk temperature profile causing viscosity changes [Hameury et al., 2009]. This is an example of an 'internal' event, which means the extreme variability we observe in some quasars is a result of significant physical changes within the accretion disk. It is still ambiguous however, whether an explanation/event like this is something any/all quasars can go through, or is characteristic of an extreme variable sub-population.

Tidal Disruption
Another example of an 'internal' explanation of extreme variability is tidal disruption events. This is where a star passes too closely to a dormant black hole and is torn apart by tidal forces. Part of the star's mass may accrete onto the disk, causing a temporary flare in radiation and therefore luminosity [Lawrence, 2016]. This is classified as a rare event due to the relatively low probability of it occurring. According to [Chan et al., 2019], a similar fraction of TDEs should occur in AGN as we observe galaxies host an AGN. This means to catch one happening, one would have to either wait 100,000 years or watch 100,000 galaxies[Lawrence, 2016]. Although this is a good estimate for the probability of one occurring, the precise number depends on the number density of stars within a given radius of the black hole, which of course is difficult to predict. Currently, these events are distinguished by the size of the flare and the rate at which they occur [Komossa and

Grupe, 2023]. However, progress is underway in trying to better classify how we detect an event happening, and how we can distinguish them from just 'normal' AGN flares. So a process like this may be responsible for the very sudden and large variability seen in some quasars, but cannot explain the more prolonged outburst observed in others. It also cannot be the primary driver of extreme variability due to the % of quasars seen exhibiting large amplitude changes in recent large-scale surveys [Rumbaugh et al., 2018].

Relativistic Effects

The first external factor which may cause observations of extreme variability in quasars are relativistic effects. Approximately 10% of AGN host powerful relativistic jets which emerge from the central region. These jets consist of relativistic particles, which when oriented towards the line of sight of the observer, can exhibit relativistic beaming/boosting. The enhancement of a quasar's luminosity due to this effect taking place may be one of the mechanisms which produce such large and rapid variability. So an effect such as this is a result of the way a quasar is oriented in the line of sight, rather than variations happening physically within the quasar. This may be able to explain some cases of extreme variability however due to the large number of cases we see in modern-day surveys, it is unlikely that all cases are a result of this [Rumbaugh et al., 2018].

Extinction/Absorption

A second external factor which may cause observations of extreme variability in quasars is obscuring material along the line of sight. This may, for example, be dust/gas between the quasar and observer, or clouds that are thought to surround SMBHs on a scale of $\approx$1 parsec. It is thought that the clouds within this 'obscuring' region are in fact clumpy, rather than smooth [Krolik and Begelman, 1988] [Nenkova1 et al., 2008]. It therefore may be plausible to assume that as these clumps move across the region, they partially (or fully) block light coming from more central regions. The dynamical timescale predicting significant changes within the obscuring region is approximately thousands of years. However, the timescale predicted for a single cloud to produce an occultation effect, as it moves across the central region, is a much more reasonable order of years. Some suggest that we can detect such an event via a reddening signature, however, the finer details of this variability explanation are yet to be ironed out. It would be reasonable to assume that is one of the external mechanisms driving the observed extreme variability, however again, not on the scale that we are seeing so many exhibit.

Microlensing

When observing AGN, especially at large distances, there is the possibility for an object such as a star in a foreground galaxy to 'lense' the light from the quasar and temporarily amplify or dim the observation [Hawkins and Taylor, 1997]. Lightcurves of these events exhibit smooth curves in brightness [Lawrence, 2016], making them reasonably distinguishable. Based on the density of stars and galaxies within the region between the quasar and the observer, we are also able to infer the timescale and rate in which we expect to observe these variations as $\approx$ years. This is consistent with the timescales we observe in some cases. Again, it would be reasonable to assume that is one of the external mechanisms driving the observed extreme variability, however, whether it is the main one, is still up for debate.

To summarise, ideally over the coming years, research will lead us to define the signatures

characterised by each of these internal and external mechanisms. And by doing this, we will be able to clarify whether extreme variability really is a result of substantial physical changes within quasars and whether this is unique to a sub-population.

4.2 Future Research

4.2.1 As a Project

Upon reflection, there are some aspects of the project which may not have worked so well, and in future could be improved on. Due to time constraints, there were also some avenues of the topic which have yet to be explored. The following may be of interest:

Different Variability Metrics
One of the pillars of this project was defining what was meant by variability. The standard deviation in quasar magnitude was chosen due to its measurement of dispersion around the light curve mean. This meant it would be generally more robust to questionable outliers within the dataset. As analysis followed though, it was discovered that a large portion of the sample was being distorted by either extreme outliers or lone unsupported data points. A comparison using a different approach in quantifying variations such as $(max - min)$, would therefore be interesting. Another more interesting and mathematical approach to variability is set out in [Morganson et al., 2014]. Here the statistic is built off of Equation 13 from [Schmidt et al., 2010] where we assume the magnitude difference (m) between two measurements of a quasar is Gaussian, with the form shown in Equation 14.

$$V_{i,j}(\Delta t_{i,j}) = \Delta m_{i,j} - \sqrt{\sigma_i^2 + \sigma_j^2} \tag{13}$$

$$P(m) = \frac{1}{\sqrt{2\pi(V^2(t) + \sigma^2)}} e^{-\frac{m^2}{2(V^2(t)+\sigma^2)}} \tag{14}$$

$$\sigma^2 = \sigma_j^2 + \sigma_k^2 \tag{15}$$

$P(m)$ is the normalised probability distribution of m while σ_j^2 and σ_k^2 are the statistical magnitude uncertainties for j and k quasar measurements. It then follows that σ^2 is "the statistical uncertainty on the magnitude difference m"[Morganson et al., 2014]. This characterization of variability is interesting as it should be more robust against the irregular cadence of multi-epoch light curves.

More Precise Data Removal
Of course, if there was more time for this project, the filtering of light curves would have ideally been done to at least $R_{Pan-STARRS} = 3$. This was not possible within the time frame due to the extremely large number of quasars the further down in $R_{Pan-STARRS}$ you go. A possible route could implement a more detailed machine learning algorithm, which detects outliers/anomalies within time series (for example an isolation forest [Sarkar, 2024]). What would be ideal though, is if questionable/lone data points were individually removed from the sample, instead of removing the whole light curve itself. This would

automatically add thousands of more quasars to the database which in turn will increase the accuracy of any relationships between properties and variability.

More Radio Loud Data
A key topic within the project that lacked substantial data was the relationship between radio-loudness and variability. If more radio measurements for the quasars within the project database (particularly the sub-samples) were obtained, a tighter relationship on whether highly variable quasars were also radio loud could be made.

The Structure Function
Perhaps the most interesting path that wasn't explored was the structure-function. In the past, astronomers employed this methodology to better quantify how the flux of these AGN sources changes over time by measuring the mean value of variance [Emmanoulopoulos et al., 2010]. Equation 16 shows the widely used form of the structure-function, where flux measurements $x(t)$ are separated by a given time interval τ [Emmanoulopoulos et al., 2010].

$$SF(\tau) = \left\langle [x(t) - x(t+\tau)]^2 \right\rangle \tag{16}$$

$$SF(\tau) \propto \tau^{\beta} \tag{17}$$

What is particularly of interest to researchers is the apparent slope of the SF, indicated in Equation 17. A break/turnover in this should indicate a characteristic timescale for which the quasars like to vary on. It would have been interesting to create structure functions for each of the sub-samples to analyse if there were differences between them.

4.2.2 As the Research Area

Speaking more broadly, the research area of highly variable AGN, promises to be very exciting over the coming years. It goes without saying that one of the most important jobs for determining the nature of quasar variability is simply more observations/data, both photometric and spectroscopic.

The Large Synoptic Survey (LSST) [Ivezić et al., 2019] (or Vera C. Rubin Observatory), located in northern Chile, is one of the most ambitious and exciting surveys taking place in the next few years. First light should take place within the year, while official observations are aimed to begin in $\approx$ mid-2025. This survey will be capable of observing the whole southern sky $\approx$ 800 times over an $\approx$ 10 year period. While this allows many astrophysical applications such as probing the Milky Way and Dark Matter [Ivezić et al., 2019], undoubtedly the field of quasar variability will explode with new data of individual quasar structure functions and their (hopefully) spectroscopic followups. The survey will also be capable of catching the rare and interesting events mentioned in the previous Section.

As mentioned in [Lawrence, 2016], the perhaps more important type of observations needed for the next generation of variability studies is large-scale spectroscopic surveys which can image the sky on the timescales we think we are seeing physical and structural

changes in the Broad Line Region (BLR) (see Section 1.2.1 for a reminder of the BLR). This is $\approx$ on timescales of months to years, which we'd expect to see for microlensing events too.

To briefly review the key remaining questions of this project and the research area as a whole: does extreme variability happen to all quasars at some point in their lifetime? Or is it just a unique subset, that only experiences it due to their physical properties? Do the quasars which exhibit extreme variations, do it again? Or is it a once-in-a-lifetime event? We hope to find out soon!

5 Conclusions

To summarise the findings of this project, highly variable quasars do not seem intrinsically different to the general population, instead just lying at the tail end of the distribution. Why do we think this? First, their σs do not stand out at all as bimodal. Second, not all sub-samples exhibit significant deviations from the general population when looking at various properties such as luminosity, Eddington ratio and radio loudness. This may indicate that these 'property deviations' are in fact not specific to the population of highly variable quasars but instead, a period of evolution they may all go through.

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